Problem Set 8: Solution

DFT/FFT Conventions:

$$X_N[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_N[k] e^{j2\pi kn/N}$$

Use L_x for the number of acquired samples in $x[n]$ and N for the DFT/FFT length. If $N > L_x$, append zeros after the L_x samples (zero-padding). All spectral experiments in this problem set use complex sinusoids to isolate positive frequency components: normalize their magnitudes by L_x, not N.

1. Periodic representations of DFT/FFT

$$x[0:3] = [1,2,0,-1]$$

(a) Compute the four point DFT by hand. Show the arithmetic and intermediate terms. Use an FFT only to verify your result.

(b) Extend the DFT formula to any integer k (i.e., including k outside the $0 \ldots N-1$ range). Show that $X[k+4]$ equals $X[k]$. What frequencies do bins $k = 3$ and $k = -1$ represent relative to the sampling frequency f_s?

(c) Evaluate the inverse DFT expression at $n = -1, 4, 5$ and 6. Show that this expression repeats every four samples. Distinguish this periodic representation from the finite sequence $x[n]$ taken to be zero outside $n = 0 \ldots 3$.

(d) Does an FFT change either of these periodicity properties? Explain.

$$X[k] = \sum_{n=0}^{3} x[n] e^{-j2\pi kn/4}$$

$$= x[0] + x[1] e^{-j\pi k/2} + x[2] e^{-j\pi k} + x[3] e^{-j3\pi k/2}$$

$$X[k] = x[0] + (-j)^k x[1] + (-1)^k x[2] + (j)^k \cdot x[3]$$

For $x[n] = [1, 2, 0, -1]$

$$X[k] = 1 + 2(-j)^k - (j)^k \Rightarrow X[0:3] = [2, 1-3j, 0, 1+3j]$$

(b) $$X[k+4] = x[0] + \underbrace{(-j)^{k+4}}_{=(j)^k(-j)^4 = (-j)^k} x[1] + \underbrace{(-1)^{k+4}}_{=(-1)^k(-1)^4 = (-1)^k} x[2] + \underbrace{(j)^{k+4}}_{=(j)^k(j)^4 = (j)^k} x[3]$$

$$= x[0] + (-j)^k x[1] + (-1)^k x[2] + (j)^k x[3] = X[k]$$

$X[-1] = X[-1+4] = X[3]$

$X[-1]$ and $X[3]$ correspond to frequency $-\frac{f_s}{4} = \frac{3f_s}{4}$ which are the same due to 2π periodicity

(c) Inverse DFT

$$x[n] = \frac{1}{4}\left[X[0] + (j)^n X[1] + (-1)^n X[2] + (-j)^n X[3]\right]$$

$$x[-1] = \frac{1}{4}\left[2 - j(1-3j) + j(1+3j)\right] = -1$$

$$x[4] = x[0] = \frac{1}{4}\left[2 + (1-3j) + (1+3j)\right] = 1$$

$$x[5] = x[1] = \frac{1}{4}\left[2 + j(1-3j) - j(1+3j)\right] = 2$$

$$x[6] = x[2] = \frac{1}{4}\left[2 - (1-3j) - (1+3j)\right] = 0$$

This is the periodic extension of $x[n]$. The finite sequence is $x[n] = \ldots, 0, 0, \underset{\uparrow}{1}, 2, 0, -1, 0, 0, \ldots$

$n = 0$

(d) The FFT does not change any of the periodicity properties. The FFT is just a computationally efficient way to compute the DFT when the blocklength N is the product of small prime factors.

2. A delay that wraps around

Let the eight sample record $x_8[n]$ be the sequence below and define its eight point DFT to be $X_8[k]$.

$$x_8[0:7] = [1, 2, 3, 4, 0, 0, 0, 0]$$

$$Y_8[k] = X_8[k]e^{-j2\pi 6k/8}$$

(a) Find the eight sample inverse DFT of $Y_8[k]$. Derive the circular shift relation and identify exactly which samples wrap around.

(b) Instead interpret the original signal as zero outside its nonzero support, samples 0 through 3. Delay it by six samples on the infinite time axis. List its first eight samples. Explain why this differs from part (a), although both calculations are often called a delay.

(c) Append zeros to create a 16 sample record, compute the 16-point DFT, and apply the phase ramp with denominator 16. List the complete output. What is the smallest transform length that avoids wrapping any non-zero delayed sample for this signal? Explain using its support.

(d) Does padding eliminate DFT periodicity, or does it change the period and provide space before non-zero samples would wrap? Explain.

(a) Using IDFT

$$\frac{1}{8}\sum_{k=0}^{7} X_8[k]\, e^{-j2\pi 6k/8}\, e^{j2\pi nk/8} =$$

$$= \frac{1}{8}\sum_{k=0}^{7} X_8[k]\, e^{j2\pi (n-6)k/8} = x_8[n-6]$$

n: 0 1 2 3 4 5 6 7 8 9

$x_8[n-6]$ = 0 0 0 0 0 0 1 2 3 4 0 0 0 0

+ 3 4 0 0 0 0 ← wrap around $n \bmod 8$

[3 4 0 0 0 0 1 2]

(b) The first eight samples of the linear delay are [0 0 0 0 0 0 1 2]

and they are different because there is no wrap around.

(c) With $N=16$ and $x[n]$ zero-padded to 16

$x[n-6] = [0\ 0\ 0\ 0\ 0\ 0\ 1\ 2\ 3\ 4\ 0\ 0\ 0\ 0\ 0\ 0]$

Because of the zero-padding any wrap around for $n \geq 16$ just adds zeros and thus does not affect the result

The support of $x_8[n]$ is 4. Delaying by 6 thus affects 10 positions and $N=10$ is the smallest DFT blocklength so that wrapping does not affect non-zero delayed samples.

(d) Padding does not eliminate DFT periodicity. It just creates more room before non-zero samples get affected by the modulo N operation on the indexes.

3. Circular and linear convolution

$$x = [1, 2, 3, 4], \qquad h = [1, -1, 1]$$

(a) Compute the full linear convolution by hand.

(b) Compute IFFT(FFT(x,N) times FFT(h,N)) for $N = 4$ and $N = 5$. Predict both results by folding the full linear convolution before running Python. Show the folding arithmetic.

(c) Find the minimum N that permits the FFT product to recover the full linear convolution without time aliasing. Verify at that length and at $N = 8$, including any trailing zero padding.

(d) Explain why using a longer FFT can fix circular convolution aliasing, even though a longer FFT does not necessarily separate two close tones. Identify what the padding accomplishes in each setting.

(a) $y[n] = \sum_k h[k] \cdot x[n-k]$ $\qquad h[k] = [1, -1, 1]$ (k=0)

$= h[0] \cdot x[n] + h[1] x[n-1] + h[2] x[n-2]$

```
 1  2  3  4
-1 -2 -3 -4
      1  2  3  4
-------------------
```

$y[n] = [1, 1, 2, 3, -1, 4]$ (n=0)

(b)

N = 4: 1, 1, 2, 3, −1, 4

−1, 4 ← wrap around mod N = 4

$y_4[n] = [0, 5, 2, 3]$

N = 5: 1, 1, 2, 3, −1, 4

4 ← wrap around mod N = 5

$y_5[n] = [5, 1, 2, 3, -1]$

(b) (contd.)

```
# N1 = 4 FFT convolution
N1 = 4
y4 = np.fft.irfft(np.fft.rfft(x, n=N1) * np.fft.rfft(h, n=N1), n=N1)
print(y4)
```

```
[0. 5. 2. 3.]
```

```
# N2 = 5 FFT convolution
N2 = 5
y5 = np.fft.irfft(np.fft.rfft(x, n=N2) * np.fft.rfft(h, n=N2), n=N2)
print(y5)
```

```
[ 5.  1.  2.  3. -1.]
```

(c) The minimum N is $N = L_x + L_h - 1 = 4 + 3 - 1 = 6$.

```
# N3 = 6 FFT convolution, minimum N that results in full linear convolution
N3 = 6
y6 = np.fft.irfft(np.fft.rfft(x, n=N3) * np.fft.rfft(h, n=N3), n=N3)
print(y6)
```

```
[ 1.  1.  2.  3. -1.  4.]
```

```
# N4 = 8 FFT convolution, full length linear convolution using power of 2 FFT
N4 = 8
y8 = np.fft.irfft(np.fft.rfft(x, n=N4) * np.fft.rfft(h, n=N4), n=N4)
y8[np.abs(y8) <= 1e-14] = 0.0   # zero out round-off noise
print(y8)
```

```
[ 1.  1.  2.  3. -1.  4.  0.  0.]
```

(d) Padding prevents the nonzero output support of the (linear) convolution from overlapping with its periodic copies. In a spectral display, padding samples an unchanged DTFT more finely (better frequency grid resolution for plotting). Thus, the two paddings serve different roles. To obtain better resolution to separate close tones, more actual data would have to be fed into the DFT/FFT, not more zero paddings.

4. Boundaries and spectral leakage

Use $f_s = 1024$ samples per second and acquire $L_x = 64$ complex samples. Consider two signals, each separately:

$$x_f[n] = e^{j2\pi f \ /f_s}, \qquad 0 \le n < 64 .$$

Use $f = 128$ Hz for the first signal and $f = 136$ Hz for the second signal. Use a rectangular observation window unless part (d) specifies otherwise.

(a) Derive the finite record DTFT as a geometric sum. Express its magnitude as a ratio of sine functions, including the limiting value at the tone frequency. State where the 64-point DFT samples this DTFT.

(b) Compute the number of cycles of the two signals in each 64 sample observation interval. Evaluate the underlying complex sinusoid at $n = 64$ and compare with $x[0]$. Which sinusoid matches the continuation implied by repeating the record periodically? Note that the matching condition is $x[64] = x[0]$ for the underlying tone, not equality of the last stored sample $x[63]$ and $x[0]$. Explain why.

(c) Plot DFT magnitudes for $N = 64$ and $N = 1024$ on the same frequency axes for each tone. Use magnitude in dB relative to 1, normalized by L_x, and a display floor of -60 dB. Explain why the unpadded 128 Hz case has only one nonzero DFT bin, yet its zero-padded plot reveals sidelobes between the original DFT sample frequencies. Did padding create these lobes?

(d) For the 136 Hz tone compare rectangular and periodic (as opposed to symmetric) Hann windows on a dense (frequency) grid. Normalize by each window sum. Explain the changes in sidelobes and main lobe width. Can tapering generally remove all leakage while retaining a rectangular window's resolving power?

(a) $$X_f(e^{j\omega}) = \sum_{n=0}^{L_x-1} e^{j2\pi f n/f_s} e^{-j\omega n} = \sum_{n=0}^{L_x-1} e^{j(\omega_f-\omega)n}$$

$\omega_f = 2\pi f/f_s$

$$= \frac{1-e^{j(\omega_f-\omega)L_x}}{1-e^{j(\omega_f-\omega)}} = \frac{e^{-j(\omega_f-\omega)L_x/2} - e^{j(\omega_f-\omega)L_x/2}}{e^{-j(\omega_f-\omega)/2} - e^{j(\omega_f-\omega)/2}} e^{j(\omega_f-\omega)(L_x-1)/2}$$

$$= \frac{\sin((\omega_f-\omega)L_x/2)}{\sin((\omega_f-\omega)/2)} e^{j(\omega_f-\omega)(L_x-1)/2}$$

$\omega_f = 2\pi f/f_s$

(a)(contd.)

$$X_f(e^{j\omega})\Big|_{\omega=\omega_f} = \sum_{n=0}^{L_x-1} \underbrace{e^{j(\omega_f-\omega_f)n}}_{=1} = L_x$$

DFT:

$$X_f[k] = X_f(e^{j\omega})\Big|_{\omega=\frac{2\pi k}{L_x}} = \frac{\sin\left(\left(\frac{2\pi k}{L_x}-\omega_f\right)L_x/2\right)}{\sin\left(\left(\frac{2\pi k}{L_x}-\omega_f\right)/2\right)} e^{-j\left(\frac{2\pi k}{L_x}-\omega_f\right)(L_x-1)/2}$$

(b) $f = 128$ Hz $\Rightarrow \omega_f = 2\pi\frac{128}{1024} \Rightarrow N_{128} = \frac{64}{1024/128} = 8$ cycles in $L_x = 64$

lands in bin 8 of DFT

$f = 136$ Hz $\Rightarrow \omega_f = 2\pi\frac{136}{1024} \Rightarrow N_{136} = \frac{64}{1024/136} = 8.5$ cycles in $L_x = 64$ samples

lands between bins 8 and 9 of DFT

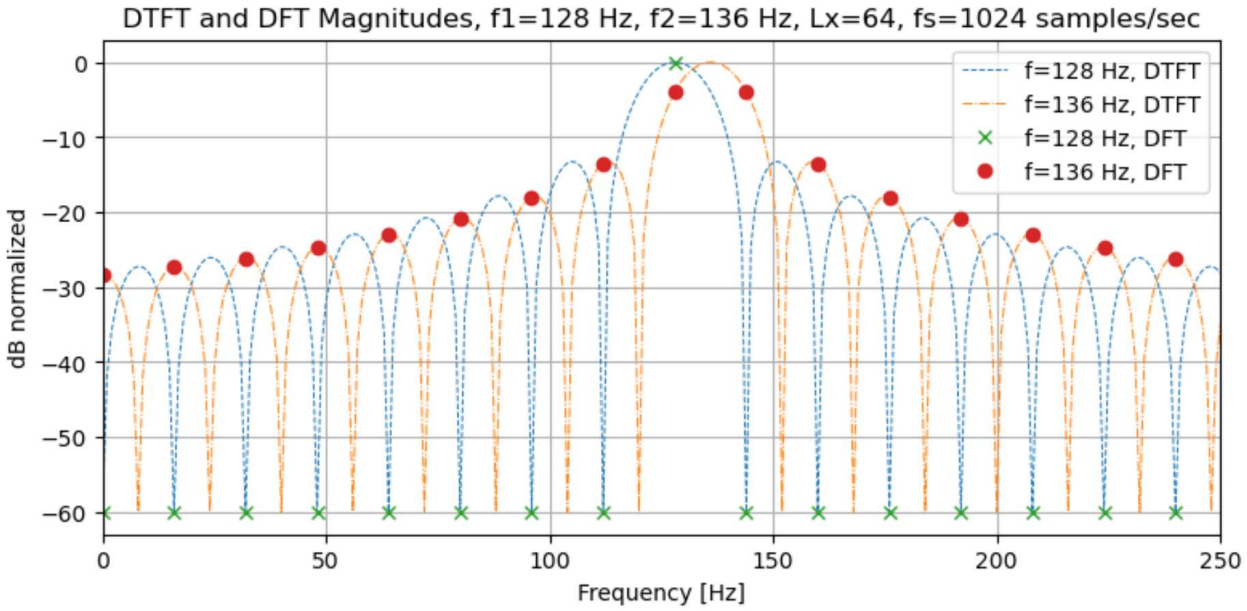
$$x_f[0] = 1, \quad x_f[L_x] = e^{j2\pi f L_x/f_s} = e^{j2\pi f/16}$$

$f = 128 \Rightarrow x_{128}[64] = e^{j2\pi\times 8} = 1$ meets continuation condition

$f = 136 \Rightarrow x_{136}[64] = e^{j2\pi\times 8.5} = -1$ does not meet continuation condition

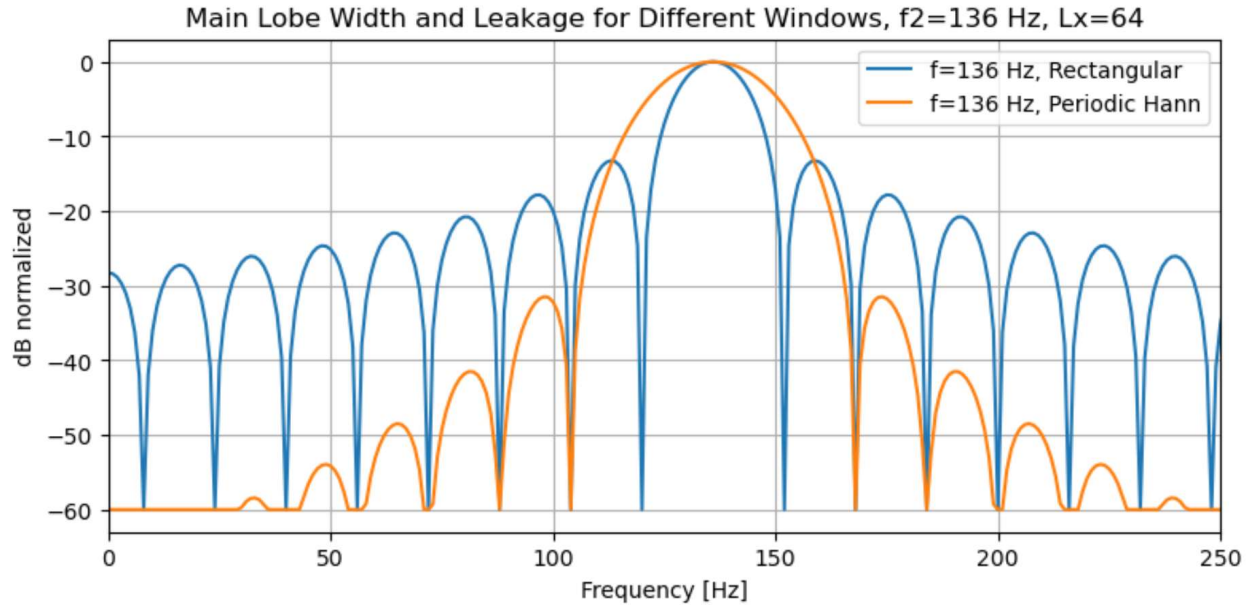
Adjacent samples of a sinusoid usually differ, so requiring the last stored sample to equal the first would be the wrong condition.

(c) DTFT (same as DFT with $N \gg L_x$) and DFT (with $N = L_x$) Magnitudes in dB



Padding does not create the sidelobes (they are a consequence of limiting the $x[n]$ sequences to $L_x = 64$). But padding makes the sidelobes visible (as $N \to \infty$, the DFT becomes the DTFT). For $f = 128$ Hz (8 cycles in $L_x = 64$) the $N = L_x$ DFT samples all sidelobes at their zero crossings, except for bin 8. For $f = 136$ Hz (8.5 cycles in $L_x = 64$) the $N = L_x$ DFT samples all sidelobes at their midpoint, except for bins 8 and 9 which are the samples of the main lobe.

(d) Rectangular and Hann windows for $f = 136$ Hz sequence with $L_x = 64$.



(d) (contd.) Compared with the rectangular window, the Hann window lowers sidelobes and doubles the approximate null to null main lobe width from 32 to 64 Hz. Leakage suppression and tone separation compete and you can not have high spectral resolution and low leakage (sidelobes) simultaneously.

5. Zero padding and frequency plot detail (ECEN 5632 Problem)

Acquire a single complex-valued sinusoid with amplitude 1, $f_s = 2048$ samples per second, $f_0 = 403.3$ Hz and $L_x = 128$ samples. Keep these 128 measured samples fixed throughout the problem.

$$x[n] = e^{j2\pi f_0 n/f_s}, \qquad 0 \le n < 128.$$

(a) Compute spectra for $N = 128, 512$ and 4096. Overlay or compare plots over 350 to 450 Hz using magnitude divided by 128. Make a table of N, frequency grid spacing, largest bin frequency in the 350 to 450 Hz interval, and its amplitude. Explain the improvement in locating the sampled peak.

(b) Write the M-point DFT of the zero padded record. Prove that it samples the same finite record DTFT at a denser set of frequencies. If $M = qL_x$ for integer q, identify exactly which padded DFT bins equal the original L_x-point DFT bins. Verify the result for $q = 4$.

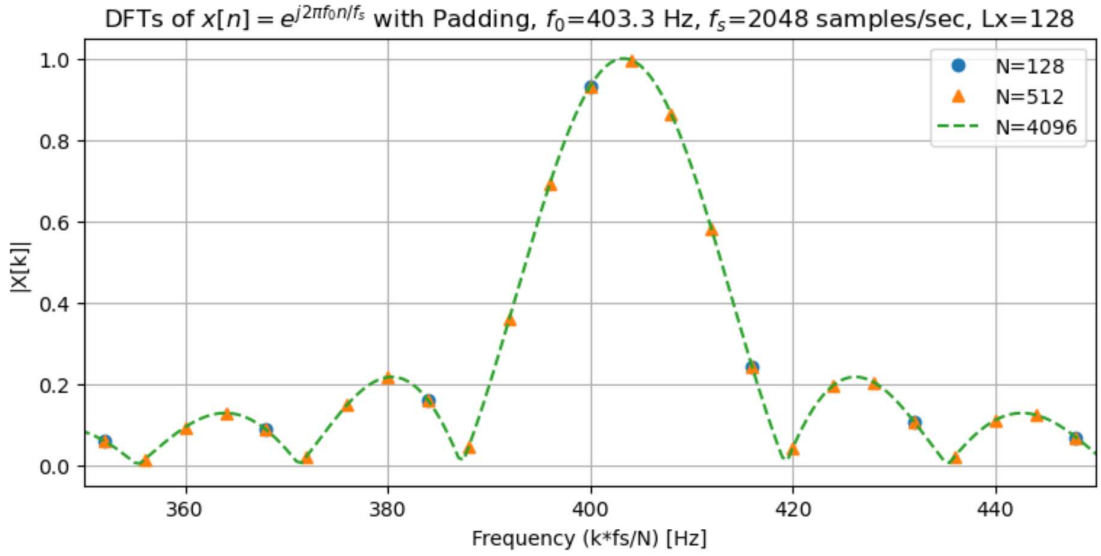
(c) Find the first spectral zeros on either side of f_0 and the null-to-null main lobe width. Do these frequencies change as N increases? Explain the distinction between plot grid spacing and main lobe width.

(d) Use the L_x-point inverse DFT to show that every M-point zero padded value can be calculated from the original L_xDFT values. Give an explicit interpolation sum. Explain why this establishes that padding supplies no additional measured information.

(e) Explain why normalization by N would falsely suggest that the signal amplitude decreases with padding. Explain why improved single tone peak localization does not establish improved two tone separation.

Note: Use $N = 128, 512$ and 4096 only as computational and display choices; do not acquire additional signal samples in this problem.

(a) Spectra for$N = 128, 512$ and 4096.



N	Grid [Hz]	Largest Bin [Hz]	Peak Amplitude
128	16.0	400.0	0.9315
512	4.0	404.0	0.9969
4096	0.5	403.5	0.9997

(b) M-point DFT of zero padded record of length $L = L_x$

$$X_M[k] = \sum_{n=0}^{L-1} x[n] e^{-j2\pi kn/M} = X_L(e^{j2\pi k/M}) \leftarrow \text{sampled DTFT}$$

$$= \sum_{n=0}^{L-1} \underbrace{\frac{1}{L} \sum_{r=0}^{L-1} X_L[r] e^{j2\pi rn/L}}_{= x[n] \text{ (IDFT)}} e^{-j2\pi kn/M}$$

$$= \frac{1}{L} \sum_{r=0}^{L-1} X_L[r] \underbrace{\sum_{n=0}^{L-1} e^{j2\pi n(r/L - k/M)}}_{\text{interpolation between } X_L[r],\ r=0,1,\ldots,L-1} \qquad k = 0, 1, \ldots, M-1$$

(b) (contd.) Note that

$$\sum_{n=0}^{L-1} e^{j2\pi n(r/L - k/M)} = \frac{\sin(\pi(r/L - k/M)L)}{\sin(\pi(r/L - k/M))} e^{j\pi(r/L - k/M)(L-1)}$$

$\sin(\pi(r/L - k/M))$: $= L$ if $r/L = k/M$

$e^{j\pi(r/L - k/M)(L-1)}$: delay term

Thus if $M = qL$, q integer, in

$$X_M[k] = \frac{1}{L} \sum_{r=0}^{L-1} X_L[r] \sum_{n=0}^{L-1} e^{j2\pi n(r/L - k/M)}$$

then $r/L - k/M = (r - k/q)/L$ and if $r = k/q$ or $k = qr$

$$X_M[qr] = X_L[r]$$

The qr indexed padded DFT bins are equal to the original r indexed bins

This can be easily seen in the graph in part (a) for $q = 4$ ($M = 4 \times L = 4 \times 128 = 512$)

(c) Main lobe width

N	left zero [Hz]	right zero [Hz]	width [Hz]
128	384.0	416.0	32.0
512	388.0	420.0	32.0
4096	387.5	419.5	32.0

The full main lobe width remains 32.0 Hz at every padding length. The main lobe width is equal to $2f_s/L_x = 4096/128 = 32$ Hz

(d) From part (b)

$$X_M[k] = \frac{1}{L}\sum_{n=0}^{L-1} X_L[r]\,\frac{\sin\left(\pi\left(\frac{r}{L}-\frac{k}{M}\right)L\right)}{\sin\left(\pi\left(\frac{r}{L}-\frac{k}{M}\right)\right)}\,e^{-j\pi\left(\frac{r}{L}-\frac{k}{M}\right)(L-1)}$$

$X_M[k]$: padded DFT; $X_L[r]$: original DFT; interpolation; delay term

The original DFT (blocklength $L = L_x$) determines all values of $X_M[k] \Rightarrow$ no new information is added to improve two-tone separation.

(e) Dividing the (padded) FFT by N introduces a false L/N amplitude reduction caused by the $N-L$ padding zeros. For a general window use the window sum instead of L.

Better single tone peak localization is a denser numerical evaluation of existing information. It is not the addition of measured observations and does not narrow the spectral lobes of two tones.